Lesson plan · for the teacher

Code on paper

Programming · Level 4 · Lesson 3

Time
40 min

Materials
Printed worksheets or the board; Paper strips and scissors; One phone to check the answers

Learning goals

- Use Parsons strips, trace tables and desk checks.
- Assess programming fairly on paper.
- Use Darsi's printed worksheets.

Activity: Paper first

1. Warm-up (5 min): Write the while loop that adds 1 to 3 on the board. Pairs write its output on paper.
2. Trace (10 min): Pupils draw a table with columns n and total, and fill one row per turn. Check the first two rows together.
3. Strips (10 min): Pairs order and indent a set of Parsons strips, with one distractor to reject.
4. Desk check (10 min): Pairs swap their trace tables and strips and check each other's, circling any disagreement.
5. Wrap-up (5 min): Run the programs on one phone and settle the disagreements. Then give the exit ticket.

This lesson works with no power and no devices; the phone at the end is optional.

Discussion

- What did the trace tables show that running the code wouldn't?
- How could you assess your next unit on paper?
- Which pupils benefit most from paper-first lessons?

Exit ticket

- Write one paper question that tests loops, and its answer.

Try it at school tomorrow
Print (or copy onto the board) questions 2 to 5 of this lesson's worksheet: the strips, the trace table, the gaps and the bug. Pairs do them on paper; at the end, one phone checks the answers.

Practice questions · for you

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1 **Half your class has no phone at home. Which homework is fairest?**

◯ A printed trace table and Parsons strips ◯ Type and run a program at home ◯ Watch a video online

2 **Paper strips for a class: the program adds 1, 2 and 3 and prints Total: 6. One strip doesn't belong.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `for n in range(1, 4):`

☐ `for n in range(1, 3):`

☐ `print("Total:", total)`

☐ `total = 0`

☐ `total = total + n`

3 **A row of the trace table. After the loop has run twice, what are n and total?**

```
n = 1
total = 0
while n <= 3:
    total = total + n
    n = n + 1
```

◯ n = 3, total = 6 ◯ n = 3, total = 3 ◯ n = 2, total = 3 ◯ n = 2, total = 2

4 **A gap-fill for paper: the program counts the marks of 10 or more.**

```
marks = [12, 7, 15]
passed = 0
for m in marks:
    if m _____ 10:
        passed = _____
print(passed)
```

Word bank: `passed + 1` `m + 1` `>=` `=`

5 **A desk check: this loop should add 1 to 3, but it never stops. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  n = 1
2  total = 0
3  while n <= 3:
4      total = total + n
5      n = n - 1
6  print(total)
```

✎ **Which part of your next programming unit could be done entirely on paper? Which part really needs a device, and why?**

Check your answers

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1 **A printed trace table and Parsons strips**

Paper exercises test the same understanding, and every pupil can do them.

2 **Top to bottom: 2, X, 4, 1, 3**

```
total = 0
for n in range(1, 4):
    total = total + n
print("Total:", total)
```

range(1, 4) gives 1, 2 and 3; range(1, 3) would stop at 2.

3 **n = 3, total = 3**

First time: total 1, n 2. Second time: total 3, n 3.

4

```
marks = [12, 7, 15]
passed = 0
for m in marks:
    if m >= 10:
        passed = passed + 1
print(passed)
```

>= includes 10; passed + 1 adds one to the count.

5 **Line 5: `n = n + 1`**

Tracing by hand shows n going 1, 0, -1…: it never passes 3, so the loop never ends.